An Exact Arithmetic Profile of $X_0(137)$

Fricke Decomposition, Newform Orbits, Period Maps, and Dedekind Data

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Abstract

We determine the exact arithmetic profile of the modular curve $X_0(137)$. The curve has genus 11; the Fricke involution acts on its weight-two cusp space with eigenspace dimensions $(4, 7)$ and trace -3, and the quotient curve has genus 4. The class group of discriminant -548 is cyclic of order 8, generated by the form $(3, 2, 46)$, whose full power cycle we display. The weight-two newspace consists of two Galois orbits of degrees 4 and 7. We give the minimal polynomials of the Hecke eigenvalue a_2 for both orbits and distinguish them from the coefficient-field defining polynomials recorded in the LMFDB. An exact rational period-map computation shows that the winding element has nonzero image for the degree-seven orbit, proving analytic rank zero for each of its conjugates. For the degree-four orbit the central value vanishes exactly; numerical evaluation of the first derivative suggests rank one but is not used as a theorem.

We also study the subspaces $V^{\pm}$ of cusp forms vanishing at both cusps, derive their cross-Hom dimension, and use its value 18 at level 137 as a worked inverse problem. Finally, we compute the classical Dedekind sums $s(h, 137)$, determine their zero set $\{37, 100\}$, and reduce five ratio classifications to residue-class quadratics using Dedekind reciprocity. All theorem-level computations use exact arithmetic; the underlying finite data are displayed in the text or described in the appendix.

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1 Introduction

The modular curve $X_0(137)$ provides a compact setting in which several classical arithmetic structures can be determined exactly and compared in a single normalization. We study its genus and Fricke decomposition, the proper class group of discriminant -548, the Galois orbits in its weight-two newspace, and the classical Dedekind sums at modulus 137. Two points require particular care. First, a polynomial defining a Hecke field need not be the minimal polynomial of the specified coefficient a_2. Second, numerical evidence for a derivative of an L-function is not an exact rank certificate. Keeping these distinctions explicit yields a sharp arithmetic profile without enlarging the theorem set beyond the available proofs.

The Fricke decomposition also gives two naturally different Hom spaces. The full eigenspaces and the cusp-vanishing hyperplanes satisfy

$$\dim \mathrm{Hom}(S_p^+, S_p^-) = d_+(p)d_-(p), \qquad \dim \mathrm{Hom}(V^+, V^-) = (d_+(p) - 1)(d_-(p) - 1).$$

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We prove the second formula for prime level and determine the primes for which its value equals the target 18 read from level 137. This is a worked inverse problem inside the stated modular-curve family, not an independent prediction of the target or a preferred level.

The Dedekind-sum results form a second exact classification. Reciprocity turns each equation $s(h_1,p)/s(h_2,p) = a/b$ into finitely many linear or quadratic equations indexed by residue classes. This proves five fixed-pair prime loci and, after an exhaustive finite calculation, the simultaneous $h \le 28$ locus. The pairs, ratios, and cutoff are hypotheses of these statements rather than invariants selected by the theory. In particular, the target fractions are taken from level 137 before asking whether other prime levels reproduce them. This illustrates an exact decision procedure; it supplies no independent explanation of the selected inputs. No physical interpretation is attached to these results.

All theorem-level computations are over exact rings or exact rational arithmetic. The larger matrix and residue-class calculations have machine-readable data and independent verification routines described in Appendix A. The only numerical statement retained in the discussion is the suggested rank-one behavior of the degree-four newform orbit, which remains explicitly conditional.

Section 2 fixes conventions and the genus bound. Sections 3–7 treat the Fricke, class-group, newform, cusp-vanishing, and fixed-target Hom calculations. Sections 8 and 9 give the Dedekind values and prime-locus classifications, followed by concluding questions and data availability.

2 Conventions and the prime-level genus bound

Let p be prime and write

$$X_p = X_0(p), \qquad S_p = S_2(\Gamma_0(p)).$$

For $p > 3$, the curve has two cusps, denoted 0 and ∞. Let w_p be the Fricke involution on X_p, and let $W_p = w_p^*$ be its pullback on

$$S_p \simeq H^0(X_p, K_{X_p});$$

the weight-two differential identification is standard [Ste07, Chapter 3, especially p. 36]. We use

$$S_p^{\varepsilon} = \ker(W_p - \varepsilon), \qquad d_\varepsilon(p) = \dim_{\mathbf{C}} S_p^{\varepsilon}, \qquad \varepsilon \in \{+1, -1\}.$$

On the upper half-plane this convention is

$$(W_p f)(z) = \frac{1}{pz^2} f\left(-\frac{1}{pz}\right), \qquad w_p^*(f(z)\,dz) = (W_p f)(z)\,dz. \tag{1}$$

The modular-symbol star sign is distinct from the Fricke eigenvalue $W_p = \pm 1$ used below. We determine d_+ and d_- as the nullities of $W_p - I$ and $W_p + I$ on the cuspidal modular-symbol realization, so the signs in the tables are fixed by the operator itself rather than by an interface label.

For prime $p > 3$, the exact genus formula used in the finite-locus proof is

$$g(X_p) = \frac{p+1}{12} - \frac{e_2(p)}{4} - \frac{e_3(p)}{3}, \qquad e_2(p) = 1 + \left(\frac{-1}{p}\right), \quad e_3(p) = 1 + \left(\frac{-3}{p}\right). \tag{2}$$

This is the prime specialization of the general $X_0(N)$ formula recorded by Ogg [Ogg74, p. 455]. Since $e_2, e_3 \in \{0, 2\}$,

$$g(X_p) \ge \frac{p+1}{12} - \frac{1}{2} - \frac{2}{3} = \frac{p-13}{12}. \tag{3}$$

For later transparency, write n_p for the number of fixed points of w_p. Riemann–Hurwitz and the identification of invariant differentials with differentials on the quotient give

$$\operatorname{tr}(W_p) = 1 - \frac{n_p}{2}, \qquad d_+(p) = \frac{g(X_p) + \operatorname{tr}(W_p)}{2}, \qquad d_-(p) = \frac{g(X_p) - \operatorname{tr}(W_p)}{2}. \tag{4}$$

Indeed, if $Y = X_p/\langle w_p \rangle$, then $2g(X_p) - 2 = 2(2g(Y) - 2) + n_p$, while $g(Y) = d_+(p)$.

3 Genus and Fricke decompositions

Theorem 3.1 (Genus and weight-two dimension at level 137)**.** *For $X = X_0(137)$,*

$$g(X) = 11, \qquad \dim_{\mathbf{C}} S_2(\Gamma_0(137)) = 11.$$

Proof. In (2), $e_2(137) = 2$ and $e_3(137) = 0$, giving $g = 11$. The weight-two differential identification gives $\dim_{\mathbf{C}} S_2(\Gamma_0(137)) = g = 11$. □

Theorem 3.2 (Fricke data at level 137)**.** *For $X = X_0(137)$, the Fricke eigenspace dimensions, trace, and quotient genus are*

$$(d_+, d_-) = (4, 7), \qquad \operatorname{tr}(W_{137}) = -3, \qquad g(X_0(137)/\langle w_{137} \rangle) = 4.$$

Proof. Ogg's fixed-point/class-number formula gives $n_{137} = 8$ [Ogg74, p. 454]. Equations (4) and Theorem 3.1 give $\operatorname{tr}(W_{137}) = -3$ and $(d_+, d_-) = (4, 7)$; Riemann–Hurwitz gives quotient genus 4 [The26].

For completeness, invariant differentials really do descend through the branched quotient. Away from the fixed points this is ordinary etale descent. At a fixed point choose a local coordinate z with $w_{137}(z) = -z$. If $f(z)\,dz$ is invariant, then $f(z) = -f(-z)$, so $f(z) = zg(z^2)$. With $t = z^2$,

$$f(z)\,dz = \frac{1}{2} g(t)\,dt,$$

which is the pullback of a holomorphic differential in the quotient coordinate. Hence the invariant differential space has dimension equal to the quotient genus.

Independently, exact modular-symbol calculation gives an involution with characteristic polynomial

$$\det(TI - W_{137}) = (T - 1)^4 (T + 1)^7.$$

This reproduces the split and trace directly in the convention (1). □

Proposition 3.3 (Neighboring genus-11 rows)**.** *The three primes* 131, 137, *and* 139 *all give genus* 11*, with*

p	$d_+(p)$	$d_-(p)$	$\operatorname{tr}(W_p)$
131	1	10	-9
137	4	7	-3
139	3	8	-5

In particular, the row at 139 *is* $(3, 8)$*, not* $(4, 7)$*.*

Proof. Ogg's formula gives

$$(n_{131}, n_{137}, n_{139}) = (20, 8, 12).$$

Equation (4) gives traces $(-9, -3, -5)$ and hence the displayed splits. Exact modular-symbol calculation over the coefficient ring gives the nullities of $W_p - I$ and $W_p + I$. In each row $W_p^2 = I$, $d_+ + d_- = g(X_p)$, and the genus agrees with (2). The displayed trace is then $d_+ - d_-$. □

Corollary 3.4 (Full cross-Fricke Hom)**.** *At level* 137*,*

$$\dim_{\mathbf{C}} \mathrm{Hom}(S_{137}^{+}, S_{137}^{-}) = 4 \cdot 7 = 28.$$

Proof. For finite-dimensional complex vector spaces A and B, $\dim \mathrm{Hom}(A, B) = (\dim A)(\dim B)$. Apply Theorem 3.2. □

4 The class group of discriminant -548

We write $\mathrm{Cl}(-548)$ for the proper class group of primitive positive-definite binary quadratic forms of discriminant -548. This is equivalently the ideal class group of $\mathbf{Q}(\sqrt{-137})$, whose fundamental discriminant is -548.

Theorem 4.1. *The group* $\mathrm{Cl}(-548)$ *is cyclic of order* 8*. The reduced primitive form*

$$Q = (3, 2, 46)$$

has exact order 8 *and generates the group. Under the form–ideal correspondence, it is the class of* $(3, w + 2)$ *in* $\mathbf{Q}(w)$*, where* $w^2 + 137 = 0$*.*

Proof. Exact reduction and composition of forms gives the following power cycle.

n	reduced representative of Q^n
1	$(3, 2, 46)$
2	$(9, 8, 17)$
3	$(6, -2, 23)$
4	$(2, 2, 69)$
5	$(6, 2, 23)$
6	$(9, -8, 17)$
7	$(3, -2, 46)$
8	$(1, 0, 137)$

Every row has discriminant -548. The first seven rows are nonprincipal, the eighth is principal, and the eight rows are precisely the complete list of reduced primitive classes. Thus Q has exact order 8, the class number is 8, and Q generates. The displayed cycle is already an explicit check; independent ideal-class and quadratic-form computations give the same invariant factor [8]. The form and class-group algorithms are documented in [Sag26b].

For the corresponding ideal $I = (3, w + 2)$, exact ideal arithmetic gives

$$I^8 = (56 - 5w), \qquad N(I^8) = 3^8 = 6561.$$

The opposite standard choice in the form–ideal correspondence replaces this generator by its inverse; its order and the cyclic group statement are unchanged. □

5 Level-137 newforms and the coefficient-polynomial distinction

At prime level 137 and weight 2, the oldspace inherited from level 1 is zero, so the cusp space is new. For a normalized Fricke eigenform write $W_{137} f = \lambda_{137}(f) f$. We use the root-number convention

$$\epsilon(f) = -\lambda_{137}(f)$$

for normalized trivial-character newforms of this prime level. Newform orbit and Atkin–Lehner conventions are reviewed in [Ste07, Chapter 9] and [BBB+21, Sections 4.2 (p. 7), 4.5 (pp. 9–11), and 8.3 (pp. 28–29)]. The W_{137} column below records the scalar $\lambda_{137}(f)$, while ϵ is the root number under the displayed sign convention; neither column is the modular-symbol star sign.

For an orbit with conjugates f^σ, "analytic rank per conjugate" means $\operatorname{ord}_{s=1} L(f^\sigma, s)$. The order of the orbit-product L-function is the sum over conjugates. We state both when a distinction matters.

Theorem 5.1. *The level-137, weight-two, trivial-character newspace has exactly two Galois newform orbits. Their degrees, Fricke signs, and root numbers are*

orbit degree	W_{137}	ϵ
4	$+1$	-1
7	-1	$+1$.

The actual minimal polynomials of a_2 are, respectively,

$$m_{4,a_2}(x) = x^4 + 3x^3 - 4x - 1, \tag{5}$$

$$m_{7,a_2}(x) = x^7 - 10x^5 + 28x^3 + 3x^2 - 19x - 7. \tag{6}$$

The degree-seven orbit has analytic rank 0, certified by algebraic central nonvanishing. Exact central vanishing holds for the degree-four orbit.

Proof. An exact 11×11 matrix for T_2 on the sign-$+1$ cuspidal modular-symbol realization has characteristic polynomial

$$\det(xI - T_2 \mid S_2(\Gamma_0(137))) = m_{4,a_2}(x) m_{7,a_2}(x).$$

The calculation gives an integral cyclic change-of-basis matrix P, with $\det P = 16$, and the exact identity $T_2 P = PC$ for the companion matrix C of this product. The two factors are irreducible over $\mathbf{Q}$, as witnessed by irreducible reductions modulo 2 and 3, respectively. Their degrees sum to 11, and their exact Atkin–Lehner eigenvalues reproduce the $(4, 7)$ Fricke split.

Let f be a normalized representative of the degree-seven orbit, put $A_7 = A_f$, and let $e = \{0, \infty\}$ be its winding symbol. For every conjugate f^σ, the integration pairing satisfies

$$L(f^\sigma, 1) = -2\pi i \langle f^\sigma, e \rangle = -2\pi i \int_0^{i\infty} f^\sigma(z)\, dz. \tag{7}$$

This is the weight-two case of the modular-symbol special-value formula [Ste07, Equation (8.5.5)]. A rational period mapping has the same kernel as the complex period mapping [Ste07, Section 10.6]. SageMath documents this kernel property and the resulting exact central-vanishing test for simple factors [Sag26a, Sag26c]. For a simple modular abelian variety A_f, the resulting exact criterion is

$$L(A_f, 1) = 0 \quad \Longleftrightarrow \quad \Theta_f(e) = 0, \tag{8}$$

where Θ_f is any rational period mapping for the newform quotient.

In the stated rational bases, the exact 23×14 rational period map for the simple modular abelian variety A_7 sends the winding symbol e to

$$(-1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0).$$

The vector is nonzero, so (8) proves $L(A_7, 1) \neq 0$. Since $L(A_7, s) = \prod_\sigma L(f^\sigma, s)$, every conjugate central value is nonzero; each conjugate and the orbit product therefore have analytic order zero at $s = 1$. The persisted rational matrix and basis data allow the nonzero period-map image to be checked directly by exact matrix multiplication.

For the degree-four orbit, the algebraic route gives exact central vanishing for every conjugate, consistently with root number -1. □

Remark 5.2. High-precision evaluations at 96 and 160 bits give a stable nonzero first derivative at every real embedding, suggesting rank one per conjugate and orbit-product order 4. Because these evaluations carry no certified error interval, both order-one statements remain conditional. The distinction between numerical rank calculations and rigorous certification is discussed in [BBB+21, Section 9.4, pp. 36–38].

5.1 Coefficient-field polynomials are not a_2 polynomials

The following two polynomials occur in the LMFDB records for the two orbits [LMF26a, LMF26b]:

$$F_4(x) = x^4 - x^3 - 3x^2 + x + 1, \tag{9}$$

$$F_7(x) = x^7 - 10x^5 + 28x^3 - 3x^2 - 19x + 7. \tag{10}$$

They are valid defining polynomials for chosen presentations of the two Hecke fields. They are *not* the minimal polynomials of the specified coefficient a_2. The latter are (5) and (6). A field presentation and the minimal polynomial of a particular element of that field are different data, even when that element happens to generate the field. This resolves the apparent polynomial discrepancy; no coefficient-level conclusion is inferred from either presentation. Explicitly, $F_7(x) = -m_{7,a_2}(-x)$, so the displayed degree-seven field generator is the negative of the specified Hecke generator. Thus F_4, F_7 are coefficient-field defining polynomials recorded by the database, whereas m_{4,a_2}, m_{7,a_2} are the minimal polynomials of the specified Hecke coefficient a_2. The two kinds of polynomial must not be substituted for one another in an arithmetic statement.

6 The canonical cusp-vanishing construction

Let $p > 3$ be prime, let $X = X_p$, and set

$$D = (0) + (\infty), \qquad V = H^0(X, K_X - D) \subset H^0(X, K_X) = S_p.$$

Thus V consists of holomorphic differentials having an additional zero at both cusps. Since w_p exchanges 0 and ∞, V is Fricke-stable; put $V^\varepsilon = V \cap S_p^\varepsilon$.

If

$$f(q) = \sum_{n \geq 1} a_n(f) q^n \in S_p,$$

then the differential associated to f is

$$f(q)\,\frac{dq}{2\pi i q} = \frac{1}{2\pi i}(a_1(f) + a_2(f)q + \cdots)\,dq.$$

It vanishes at ∞ exactly when $a_1(f) = 0$.

Theorem 6.1 (Cusp-vanishing Fricke blocks)**.** *For every prime $p > 3$, the space V is Fricke-stable. If $d_\varepsilon(p) > 0$, then*

$$V^\varepsilon = \ker(a_1 : S_p^\varepsilon \longrightarrow \mathbf{C}), \qquad \dim V^\varepsilon = d_\varepsilon(p) - 1.$$

If both Fricke blocks are nonzero, then

$$\dim \mathrm{Hom}(V^+, V^-) = (d_+(p) - 1)(d_-(p) - 1). \tag{11}$$

At $p = 137$,

$$(\dim V^+, \dim V^-) = (3, 6), \qquad \dim \mathrm{Hom}(V^+, V^-) = 18.$$

If either Fricke block is zero, the corresponding V^ε and the cross-Hom space are zero.

Proof. Because w_p exchanges the two points of D, it preserves the line bundle $K_X - D$ and hence V. On a W_p-eigenvector, vanishing at one cusp is equivalent to vanishing at the other. The local q-expansion above therefore gives

$$V^\varepsilon = \ker(a_1 | S_p^\varepsilon).$$

It remains to show that a_1 is not the zero functional on a nonzero Fricke block. The prime-level weight-two cusp space is new because $S_2(\mathrm{SL}_2(\mathbf{Z})) = 0$. The newform decomposition is compatible with Fricke, and every nonzero constituent contains a normalized eigenform with $a_1 = 1$; equivalently, the perfect Hecke pairing and $a_1(T_n f) = a_n(f)$ give the same conclusion [Ste07, Lemma 3.22 and Proposition 3.24]. Thus the kernel has codimension one whenever the block is nonzero. Equation (11) is finite-dimensional linear algebra. Substituting $(d_+, d_-) = (4, 7)$ gives $(3, 6)$ and 18. If a block is zero, its intersection with V is zero and the cross-Hom space is zero; the bare signed product in (11) is not used in that edge case. □

Remark 6.2 (Hecke boundary). The space V is Fricke-stable, but when $V \neq 0$ it is not stable under every Hecke operator. Indeed, if every T_n preserved V, then for $f \in V$ one would have $a_n(f) = a_1(T_n f) = 0$ for every n, so the q-expansion principle would force $f = 0$. Thus the construction uses the canonical hyperplanes $\ker(a_1)$ as vector spaces. No quotient Hecke algebra, Hecke-stable reduced block, or canonical complementary line is asserted.

Remark 6.3 (Full and cusp-vanishing Hom spaces are distinct). At $p = 137$,

$$\dim \mathrm{Hom}(S_{137}^+, S_{137}^-) = 28, \qquad \dim \mathrm{Hom}(V^+, V^-) = 18.$$

The second space is defined from canonical hyperplanes inside the two Fricke blocks. There is a canonically defined restriction sequence

$$0 \longrightarrow \mathrm{Hom}(S_{137}^+ / V^+, S_{137}^-) \longrightarrow E \longrightarrow \mathrm{Hom}(V^+, V^-) \longrightarrow 0,$$

where

$$E = \{T \in \mathrm{Hom}(S_{137}^+, S_{137}^-) : T(V^+) \subseteq V^-\}.$$

The last arrow restricts a map to V^+. It is surjective because linear maps extend after a choice of complements, but no such choice is canonical. Hence the exact sequence has no selected canonical splitting: there is no asserted canonical inclusion of the 18-dimensional Hom space into the 28-dimensional one, and the two spaces are not identified.

7 A worked cross-Fricke inverse problem

We now fix the modular-curve family, weight, character, cusp divisor, Fricke convention, and numerical target, and ask which prime levels attain that target. Here that target is the already computed value at $p = 137$.

Theorem 7.1. *Fix the family $X_0(p)$ for prime p, weight 2, trivial character, the Fricke convention above, the divisor $D = (0) + (\infty)$ for $p > 3$, and the target*

$$\dim \mathrm{Hom}(V^+, V^-) = 18.$$

Then $p = 137$ is the unique prime attaining the target. For $p = 2$ and $p = 3$, the weight-two cusp space is zero and the reduced cross-Hom dimension is set to zero.

Proof. First suppose $p > 3$ and both Fricke blocks are nonzero. Put $u = d_+(p) - 1$ and $v = d_-(p) - 1$. If $uv = 18$, the positive factor pairs up to interchange are

$$(u, v) \in \{(1, 18), (2, 9), (3, 6)\}.$$

Because $g(X_p) = d_+(p) + d_-(p) = u + v + 2$, the only possible genera are

$$21, \quad 13, \quad 11.$$

The bound (3) then forces $p \leq 265$. Exact enumeration of the primes in this bounded range gives the complete list of rows having one of the three forced genera shown below. The genus and Fricke nullities are computed over exact coefficient rings. The fixed-point counts are the corresponding exact Ogg values, and (4) independently turns them into the same traces and splits.

p	$g(X_p)$	n_p	$\mathrm{tr}(W_p)$	(d_+, d_-)	$\dim \mathrm{Hom}(V^+, V^-)$
131	11	20	-9	$(1, 10)$	0
137	11	8	-3	$(4, 7)$	18
139	11	12	-5	$(3, 8)$	14
163	13	4	-1	$(6, 7)$	30
251	21	28	-13	$(4, 17)$	48
257	21	16	-7	$(7, 14)$	78

The only target row is 137. If one Fricke block is zero, the cross-Hom dimension is zero by Theorem 6.1. Finally, $g(X_0(2)) = g(X_0(3)) = 0$, so their weight-two cusp spaces and cross-Hom spaces are zero. This exhausts all primes. □

Remark 7.2. The uniqueness is relative to the stated family and the fixed target 18; the theorem does not select that target independently.

8 Classical Dedekind sums at modulus 137

Definition 8.1 (Classical Dedekind sum)**.** For an integer h and a positive integer k, define

$$s(h, k) = \sum_{j=1}^{k-1} \left(\!\left(\frac{j}{k}\right)\!\right) \left(\!\left(\frac{hj}{k}\right)\!\right),$$

where

$$((x)) = \begin{cases} x - \lfloor x \rfloor - \frac{1}{2}, & x \notin \mathbf{Z}, \\ 0, & x \in \mathbf{Z}. \end{cases}$$

The finite sum is defined without a coprimality hypothesis. Coprimality enters the inversion and reciprocity identities below, not the definition.

Proposition 8.2 (Classical identities)**.** *For integers h and positive integers k:*

1. $s(h+k,k) = s(h,k)$;
2. $s(-h,k) = -s(h,k)$;
3. *if* $\gcd(h,k) = 1$ *and* $hh' \equiv 1 \pmod{k}$, *then* $s(h',k) = s(h,k)$; *and*
4. *if* $h, k > 0$ *and* $\gcd(h,k) = 1$, *then*

$$s(h,k) + s(k,h) = -\frac{1}{4} + \frac{1}{12}\left(\frac{h}{k} + \frac{k}{h} + \frac{1}{hk}\right). \tag{12}$$

Proof. The first two assertions follow directly from the sawtooth function, and the third follows by permuting the nonzero residue classes modulo k. The fourth is the classical Dedekind reciprocity law; see [RG72, Chapter 2]. □

Proposition 8.3 (Selected exact values at modulus 137)**.** *The following values and reduced-numerator factorizations are exact.*

h	$s(h,137)$	*numerator*	h	$s(h,137)$	*numerator*
3	$495/137$	$3^2 \cdot 5 \cdot 11$	13	$45/137$	$3^2 \cdot 5$
7	$180/137$	$2^2 \cdot 3^2 \cdot 5$	14	$108/137$	$2^2 \cdot 3^3$
8	$102/137$	$2 \cdot 3 \cdot 17$	15	$18/137$	$2 \cdot 3^2$
9	$120/137$	$2^3 \cdot 3 \cdot 5$	18	$44/137$	$2^2 \cdot 11$
11	$140/137$	$2^2 \cdot 5 \cdot 7$	19	$10/137$	$2 \cdot 5$
22	$20/137$	$2^2 \cdot 5$	28	$90/137$	$2 \cdot 3^2 \cdot 5$
37	0	0	100	0	0

In particular,

$$\frac{s(22,137)}{s(7,137)} = \frac{1}{9}, \qquad \frac{s(19,137)}{s(9,137)} = \frac{1}{12}, \qquad \frac{s(13,137)}{s(7,137)} = \frac{1}{4}, \tag{13}$$

$$\frac{s(14,137)}{s(28,137)} = \frac{6}{5}, \qquad \frac{s(15,137)}{s(18,137)} = \frac{9}{22}. \tag{14}$$

The same table also gives

$$\frac{s(7,137)}{s(22,137)} = 9, \quad \frac{s(9,137)}{s(22,137)} = 6, \quad \frac{s(11,137)}{s(22,137)} = 7, \quad \frac{s(3,137)}{s(7,137)} = \frac{11}{4}.$$

Proof. Direct evaluation of the sawtooth definition in rational arithmetic agrees with the Euclidean recurrence obtained from reciprocity. Both routes give the displayed reduced fractions, factorizations, and ratios. □

Proposition 8.4 (Symmetry and zeros)**.** *For $1 \le h \le 136$,*

$$s(137-h,137) = -s(h,137), \qquad s(h^{-1} \bmod 137, 137) = s(h,137).$$

The complete zero set is

$$\{h : 1 \le h \le 136,\ s(h,137) = 0\} = \{37, 100\}.$$

Proof. The two displayed symmetries follow from Proposition 8.2, since 137 is prime. Exact enumeration of all 136 residue representatives gives the two zeros. □

9 Do other primes share the level-137 ratios?

Fix the argument pairs below and take the target ratios from their values at $p = 137$. We classify the primes that satisfy the corresponding arithmetic equations; the final theorem also fixes an explicit search window for the arguments. The mathematical mechanism is the following residue-class quadratic reduction; the selected numerical targets are examples of that decision procedure, not independently derived inputs.

Proposition 9.1 (Residue-class quadratic reduction)**.** *Let $h, p > 0$ be coprime and let r_h be the least nonnegative residue of p modulo h; when $h > 1$, it is a unit residue. Define*

$$\beta_h(r_h) = 3h + 12h\, s(r_h, h), \qquad Q_{h,r_h}(X) = X^2 - \beta_h(r_h)X + h^2 + 1.$$

Then

$$12hp\, s(h,p) = Q_{h,r_h}(p). \tag{15}$$

Consequently, for positive integers h_1, h_2, a reduced target a/b, and a prime p coprime to h_1h_2 for which $s(h_2,p) \neq 0$, the equation

$$\frac{s(h_1,p)}{s(h_2,p)} = \frac{a}{b}$$

is equivalent in each unit residue class modulo $L = \operatorname{lcm}(h_1,h_2)$ to

$$P_r(p) = bh_2 Q_{h_1,r_1}(p) - ah_1 Q_{h_2,r_2}(p) = 0, \tag{16}$$

where $r_i \equiv r \pmod{h_i}$. Each P_r has degree at most two.

Proof. Apply reciprocity (12) and periodicity to obtain (15); clearing the quotient gives (16). Primes dividing h_1h_2, which lie outside the reciprocity hypothesis, must instead be evaluated directly from Definition 8.1. □

Theorem 9.2 (Five fixed-input prime loci)**.** *For each row below, the domain is all positive primes p for which the denominator sum is nonzero. In every row, the displayed ratio equals its supplied target if and only if $p = 137$. Put*

$$Z_2 = \{p : s(h_2,p) = 0\}, \qquad Z_{00} = \{p : s(h_1,p) = s(h_2,p) = 0\},$$

and let $Z_{\neq 0,0}$ contain the denominator-zero primes having nonzero numerator. The exact classification is

$(h_1, h_2, a/b)$	*defined locus*	Z_2	Z_{00}	$Z_{\neq 0,0}$
$(22, 7, 1/9)$	$\{137\}$	$\{2,5,7\}$	$\{2,5\}$	$\{7\}$
$(19, 9, 1/12)$	$\{137\}$	$\{2,3,41\}$	$\{2\}$	$\{3,41\}$
$(13, 7, 1/4)$	$\{137\}$	$\{2,5,7\}$	$\{2,5\}$	$\{7\}$
$(14, 28, 6/5)$	$\{137\}$	$\{2,5,7,157\}$	$\{2,7\}$	$\{5,157\}$
$(15, 18, 9/22)$	$\{137\}$	$\{2,3,5,13\}$	$\{2,3,5\}$	$\{13\}$

These are complete finite sets of primes, not search cutoffs.

Proof. For the five rows, Proposition 9.1 produces respectively $60, 108, 72, 12$, and 24 unit-residue equations, 276 in total. No equation is the zero polynomial. Each exact linear or quadratic equation is solved over $\mathbf{Q}$, then filtered by its residue class, positive integrality, primality, and the nonzero-denominator condition. The only retained prime in each row is 137. Every prime dividing an input argument is evaluated directly, and a separate exact denominator equation gives the complete undefined sets printed in the table.

For the first row, the following equivalent notation makes the residue quadratic explicit. For a unit residue r modulo 154, set

$$C_r = \frac{63\beta_{22}(r \bmod 22) - 22\beta_7(r \bmod 7)}{1848}.$$

The residue equation is

$$41p^2 - 1848C_r p + 29455 = 0. \tag{17}$$

For $r = 137 \bmod 154$, the exact values $\beta_{22} = 102$, $\beta_7 = 27$, and $C_r = 243/77$ give

$$41p^2 - 5832p + 29455 = 0, \tag{18}$$

whose rational roots are 137 and $215/41$. The factorization $29455 = 5 \cdot 43 \cdot 137$ concerns the constant term, not the discriminant, and is a consequence of the integer root rather than independent selection evidence. At $p = 5$, both $s(22, 5)$ and $s(7, 5)$ vanish. Thus $p = 5$ solves only the cross-multiplied equation $0 = 0$; the quotient is undefined. □

Corollary 9.3 (The $(22, 7)$ prime locus)**.** *For every prime p with $s(7, p) \neq 0$,*

$$\frac{s(22, p)}{s(7, p)} = \frac{1}{9} \quad \Longleftrightarrow \quad p = 137.$$

Proof. This is the first row of Theorem 9.2, with its quotient domain stated explicitly. □

Theorem 9.4 (An $h \leq 28$ target window)**.** *Let*

$$\mathcal{T} = \left\{ \frac{1}{9}, \frac{1}{12}, \frac{1}{4}, \frac{6}{5}, \frac{9}{22} \right\}$$

and, for a prime p, let

$$\mathcal{R}_{28}(p) = \left\{ \frac{s(h_1, p)}{s(h_2, p)} : 1 \leq h_1, h_2 \leq 28, \ s(h_2, p) \neq 0 \right\}.$$

Then, for primes $p \geq 5$,

$$\mathcal{T} \subseteq \mathcal{R}_{28}(p) \quad \Longleftrightarrow \quad p = 137.$$

Proof. Apply Proposition 9.1 to the $5 \cdot 28 \cdot 28 = 3920$ target and ordered-pair choices. Across the unit residue classes this gives 252940 equations: 252712 quadratic, 223 linear, and five nonzero constant equations, with no identity equation. Solving these equations over $\mathbf{Q}$ gives 162 nonempty choices in their range of applicability. Including direct evaluations at the exceptional small primes $5 \leq p \leq \max(h_1, h_2)$ brings the total to 195 nonempty choices and proves the other 3725 empty. The five target-prime sets have sizes $31, 12, 40, 8$, and 7; their intersection is $\{137\}$, and the largest prime in any individual set is 11813.

These residue-class equations cover $p > \max(h_1, h_2)$. For $5 \leq p \leq \max(h_1, h_2)$, including primes dividing an argument, direct evaluation of the sawtooth definition supplies the exceptional cases. At $p = 137$, the five pairs in Theorem 9.2 are explicit witnesses. The complete coefficient data and an exact-arithmetic verification routine are described in Appendix A. □

Remark 9.5 (Interpretation). Both theorems solve arithmetic equations whose pairs and fractions are chosen from the level-137 data; the second also fixes the cutoff 28. Here "prime locus" means only the exact solution set under those inputs.

10 Conclusion

The arithmetic of $X_0(137)$ brings several exact structures into a single picture: genus 11, Fricke split $(4, 7)$, quotient genus 4, a cyclic class group of order 8, and two newform orbits of degrees 4 and 7. The equality between the Fricke fixed-point count $n_{137} = 8$ and the class number $h(-548) = 8$ places the geometric and quadratic-form calculations side by side. The rational period map proves rank zero for the degree-seven orbit, while a rigorous treatment of the degree-four derivative remains an open next step.

The Hom and Dedekind classifications illustrate a common method: an intrinsic formula first bounds the possibilities, after which exact finite arithmetic determines the locus. It would be useful to study other cross-Fricke targets uniformly and to formulate broader residue-quadratic theorems for families of Dedekind ratios. The present results supply exact test cases for both directions without claiming that the chosen targets are themselves canonical.

A Computational data

Machine-readable supplementary data record the exact 11×11 matrix of T_2, the cyclic similarity and irreducibility data, the 23×14 rational period map, and its nonzero winding-symbol image. A second dataset contains all 3920 target and ordered-pair cases and all 252940 unit-residue equations used in Theorem 9.4. Verification routines recompute the relevant matrix identities, rational roots, exceptional primes, and direct Dedekind sums using exact arithmetic. These files are included in the accompanying package: `certificates/` contains the exact data and `tools/` contains their verifiers.

The smaller modular-symbol and class-group computations were carried out in SageMath 10.7 with PARI 2.17.3. All theorem-level matrix, polynomial, form-class, dimension, Dedekind-value, and residue-root statements use exact arithmetic. The high-precision derivative evaluations for the degree-four orbit are retained only as conditional diagnostics.

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